



Introduction

Brief History of Spotify

Spotify was founded in **2006** by **Daniel Ek** and **Martin Lorentzon** in Stockholm, Sweden. The service **officially launched** in **October 2008**.

Key features and services :-

Music Streaming : over **70 million tracks**, allowing users to stream music on-demand.

Playlists: Users can **create, share, and discover playlists**, including curated playlists like "**Discover Weekly**" and "**Release Radar**".

Podcasts: Spotify hosts a wide range of **podcasts**, including exclusive content and popular shows.

Personalized Recommendations: Using advanced algorithms, Spotify provides personalized music **recommendations based on user listening habits**.

Offline Listening: Premium users can **download music and podcasts** for offline listening.

Cross-Platform Access: Spotify is available on various **devices, including smartphones, tablets, computers, smart speakers, and more**.

Social Features: Users can **follow friends, share music on social media, and see what others are listening to** in real-time.

Source and Description of the Dataset

Source :-

The dataset is typically sourced from Spotify's public API known as **Kaggle**.

Description :-

This dataset contains information on various **Spotify tracks**, including their audio features. It comprises numerous rows, each representing a unique track, with columns for various attributes such as **track name**, **artist name**, **popularity**, **genre**, and **several audio features**.

Glossary

Track_Name : The title of the song.

Track_Artist : The name of the artist or band performing the track.

Album_Name : The name of the album the track is part of.

Playlist_Name : The name of the playlist where the track is featured.

Track_Popularity : A numerical value indicating the popularity of the track on a scale from 0 to 100.

Playlist_Genre : The genre category of the track.

Valence : A measure from 0 to 1 describing the musical positiveness of the track.

Energy : A measure from 0 to 1 indicating the intensity and activity of the track.

Tempo : The speed or pace of the track, measured in beats per minute (BPM).

Liveness : A measure from 0 to 1 that detects the presence of an audience in the recording.

Acousticness : A measure from 0 to 1 indicating the likelihood that the track is acoustic.

Danceability : A measure from 0 to 1 that describes how suitable a track is for dancing.

Duration_ms : The length of the track in milliseconds.

Instrumentalness : Predicts the likelihood of a track being instrumental, with values closer to 1 indicating more instrumental content.

Track_href : A URL to the Web API endpoint providing detailed information about the track.

Speechiness : Measures the presence of spoken words in a track, with higher values indicating more speech-like content.

Time_signature : The overall time signature of a track, expressed as a notational convention (e.g., 4/4, 3/4). [No of beats in piece of music]

Liveness : Detects the presence of an audience, with higher values indicating a higher likelihood of a live performance.

Loudness : The overall loudness of a track in decibels (dB), ranging from -60 to 0 dB.

URI : The unique identifier for the track within the Spotify database.

Analysis_URL : A URL providing access to detailed audio analysis of the track.

Mode : Indicates the modality (major or minor) of a track, where major is 1 and minor is 0.

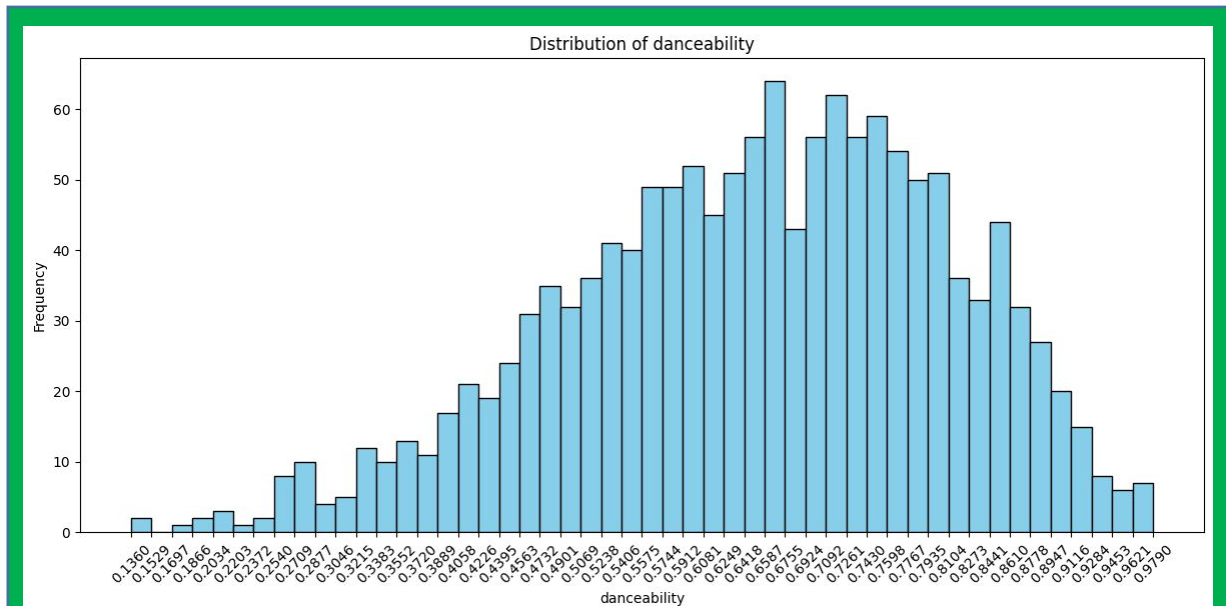
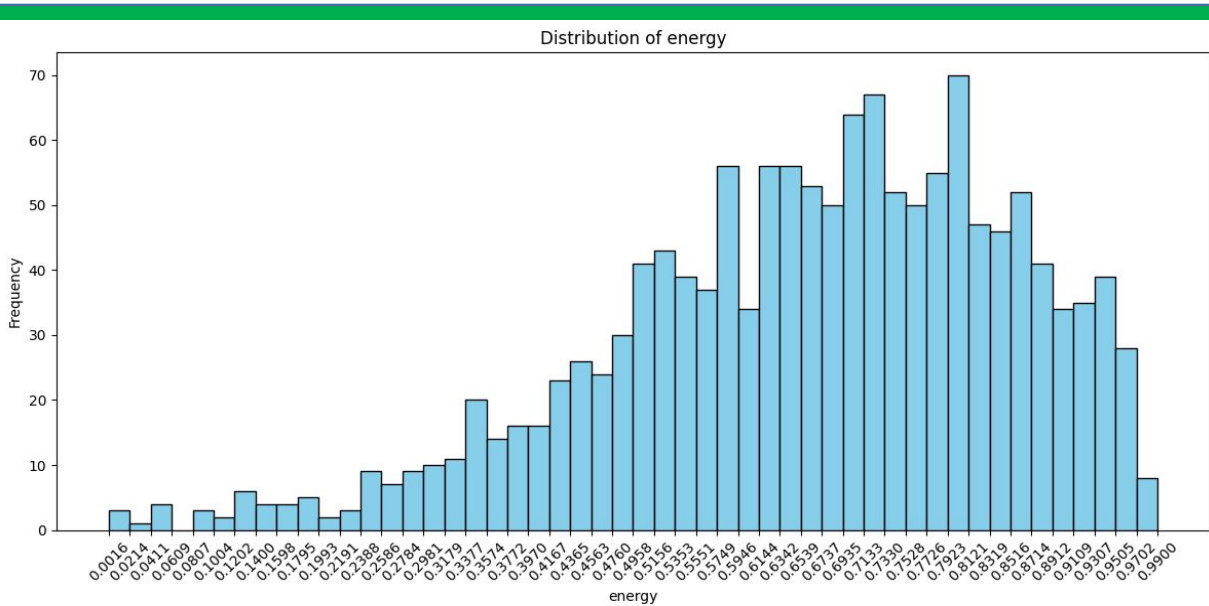
Comparisons

Energy

- Analysis indicates that **most tracks have higher energy levels**, with a **few exceptions showing significantly lower energy**.
- Tracks with higher energy levels are often associated with **high-energy** genres like **rock and pop**. They tend to be more popular on workout and party playlists. **Low-energy** tracks are typically found in genres like **ambient, classical, or indie**.
- Capitalizing on **high-energy tracks can help capture audiences** looking for vibrant and dynamic listening experiences.

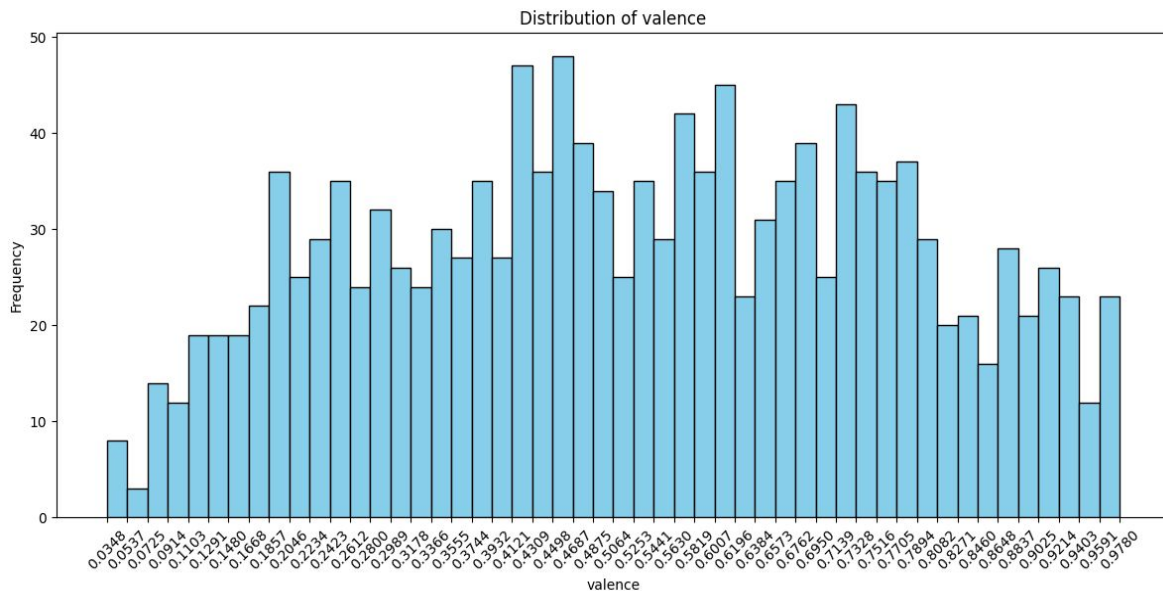
Danceability

- Analysis indicates that **most tracks have higher danceability levels**, with a **few exceptions showing significantly lower danceability**.
- **Higher danceability** scores indicate that a track is suitable for dancing. This is common in genres like **pop, dance, and electronic music**. **Lower danceability** scores might be found in genres like **classical or folk music**.
- **Curating playlists with highly danceable tracks** can attract listeners looking for music to enjoy at social events or parties.



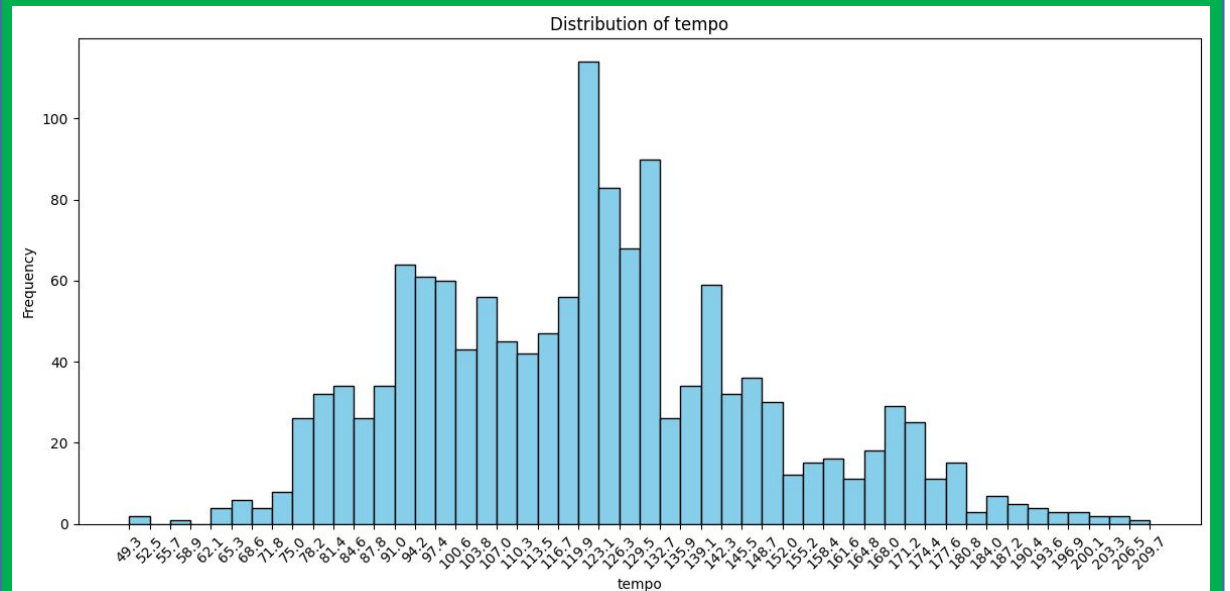
Valence

- Valence describes the musical positiveness conveyed by a track. **High valence** tracks sound more cheerful and happy e.g. **pop**, **reggae**, while **low valence** tracks sound more sad e.g., **blues**, **downtempo**.
- Understanding the valence of tracks can help in **creating playlists** that **evoke specific emotional responses**, enhancing **user experience**.



Tempo

- Tempo, measured in beats per minute (BPM), varies widely across genres. **Fast-tempo tracks** (e.g. 120-140 BPM) are common in **dance** and **upbeat pop music**, while **slower tempos** (e.g. 60-80 BPM) are found in **ballads** and **chill-out music**.
- Playlists with a variety of tempos can cater to different **moods and activities**, such as **relaxation**, **focus**, or **high-intensity workouts**.

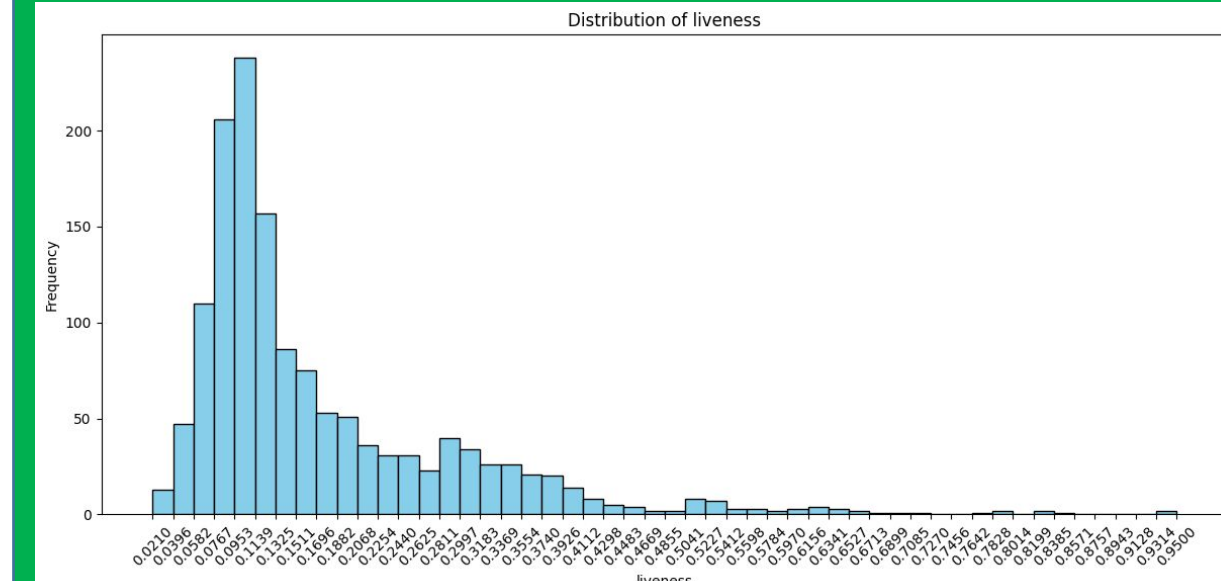
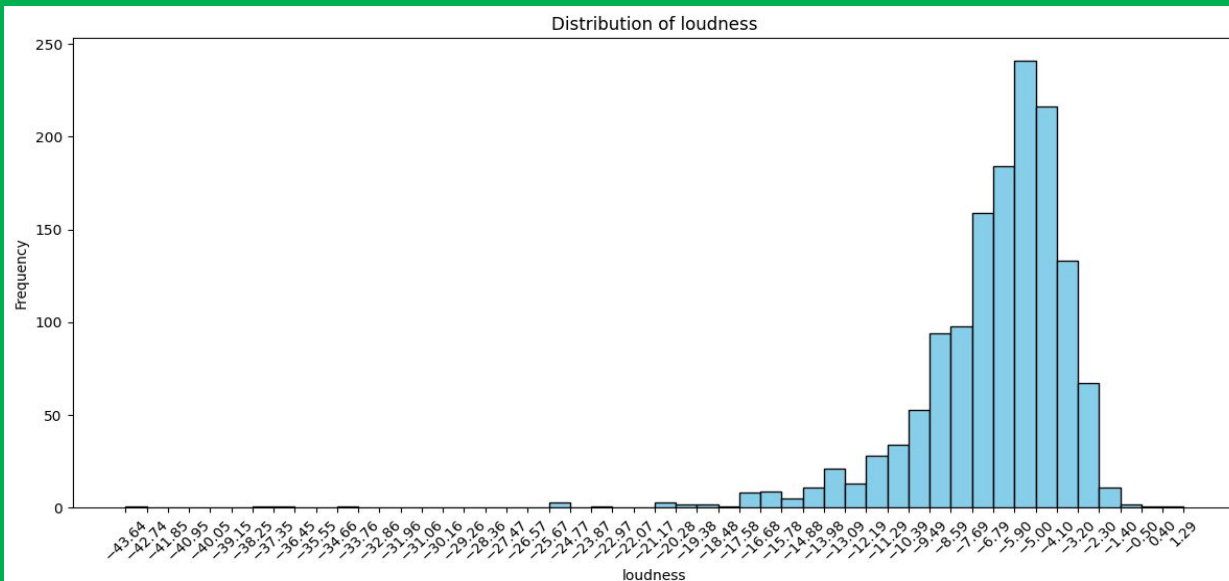


Liveness

- Higher liveness values indicate a higher probability of live performance. This is often **higher in live albums or recordings** with significant audience interaction.
- Featuring tracks with high liveness can appeal to listeners who enjoy the **live concert experience** and are looking for **authentic, energetic performances**.

Loudness

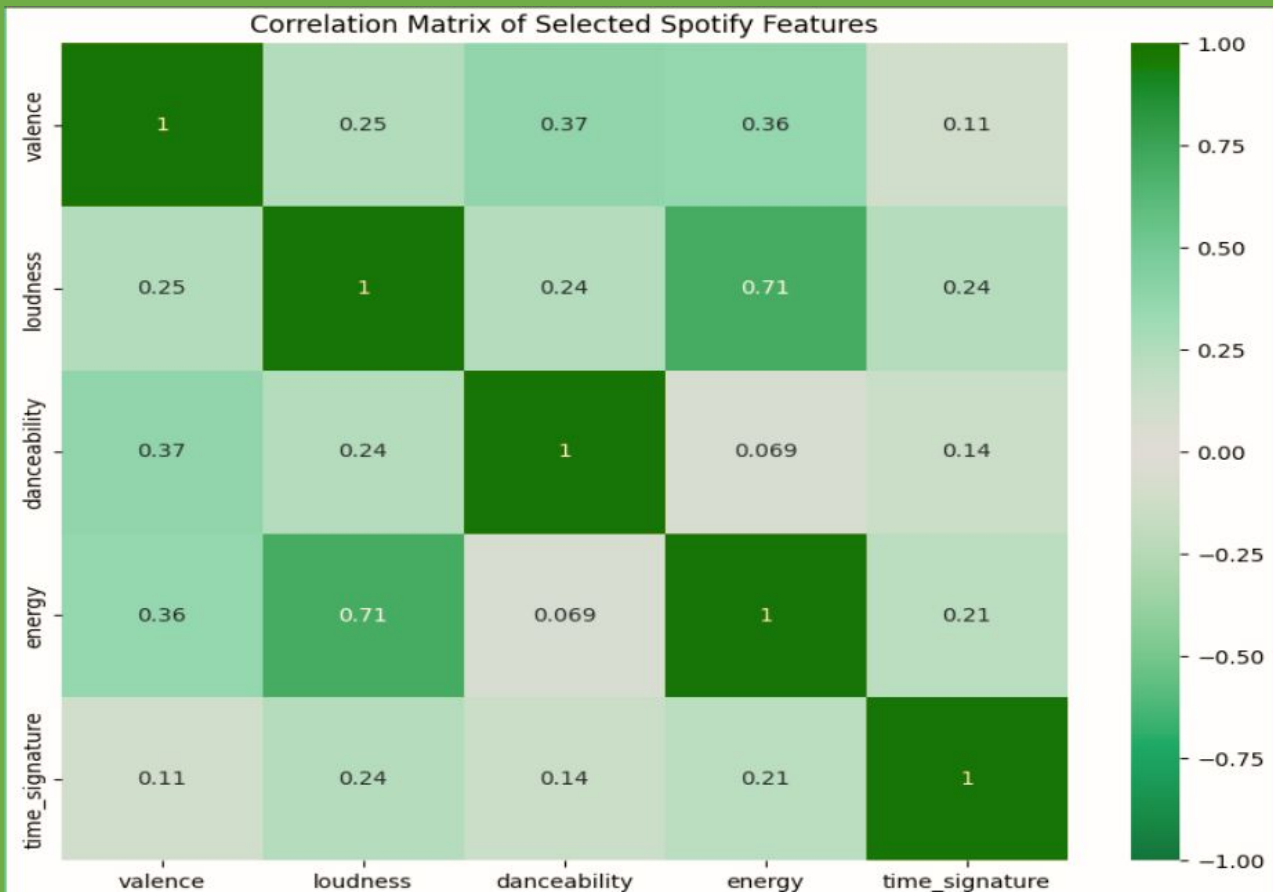
- Loudness measures the overall volume of a track. Genres like **rock, metal, and electronic** music tend to **have higher loudness**, while genres like **classical and jazz** might have lower loudness levels.
- Ensuring a **mix of loud and softer tracks** can cater to different listening environments and preferences, providing a balanced auditory experience.



Correlation

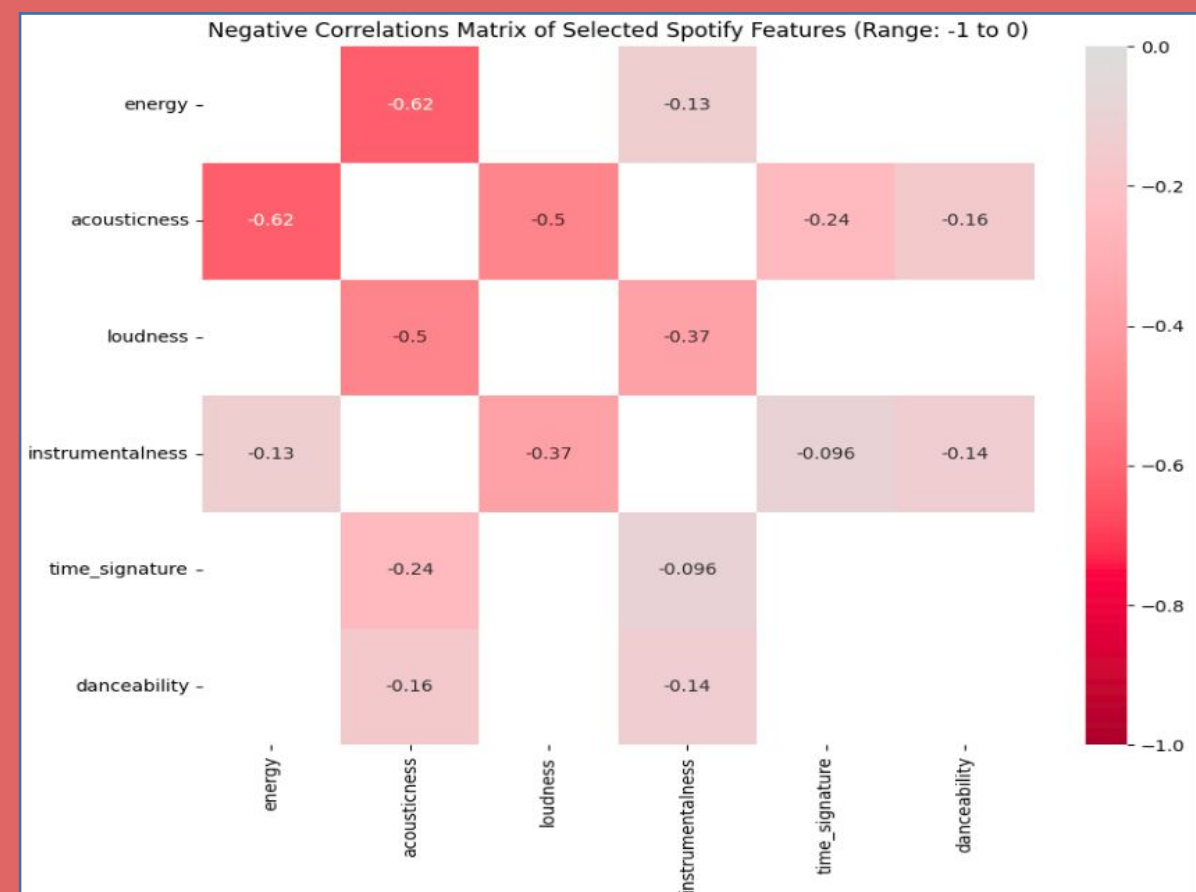
Positive correlation

- Highly energetic tracks are ideal for dancing.
- Highly loud , dancing song brings more positivity.
- Highly loud songs have more beats than any other.



Negative correlation

- Highly energetic, loud songs have very less acousticness and instrumentalness .
- Music of full of beats does not have acousticness.



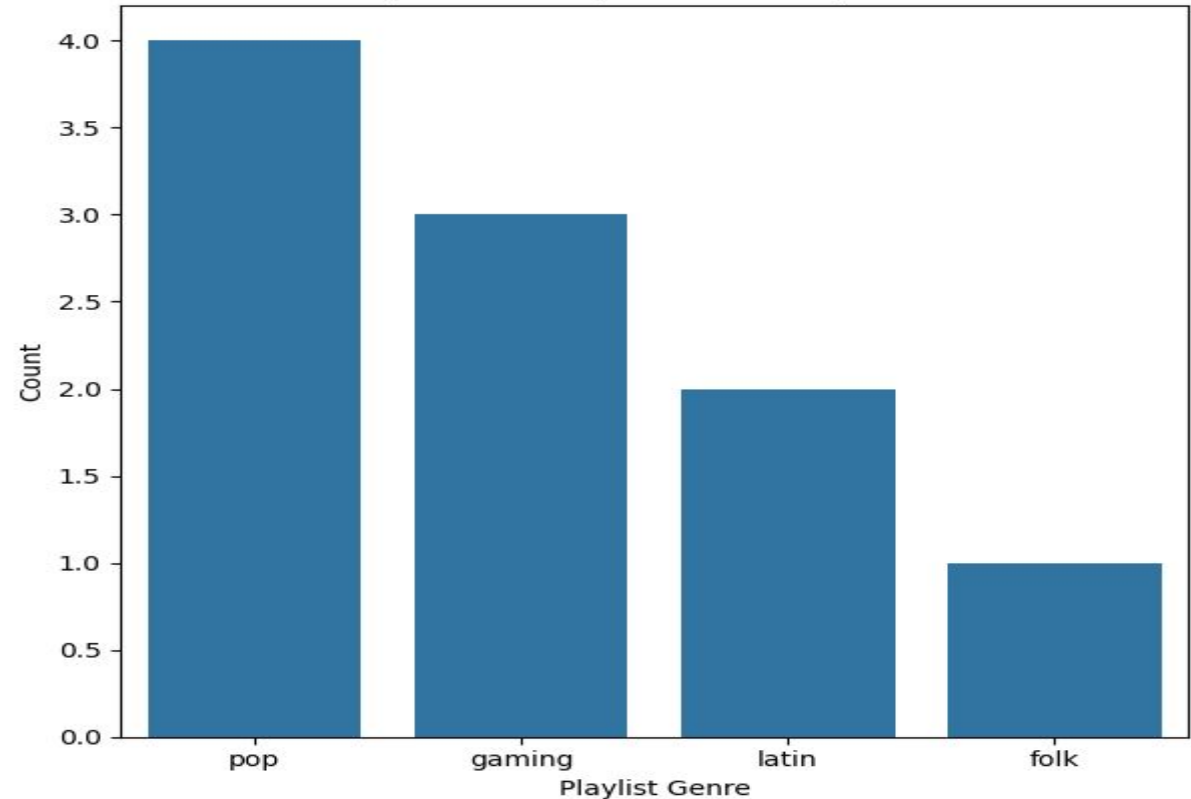
Popularity

Most Popular Playlist

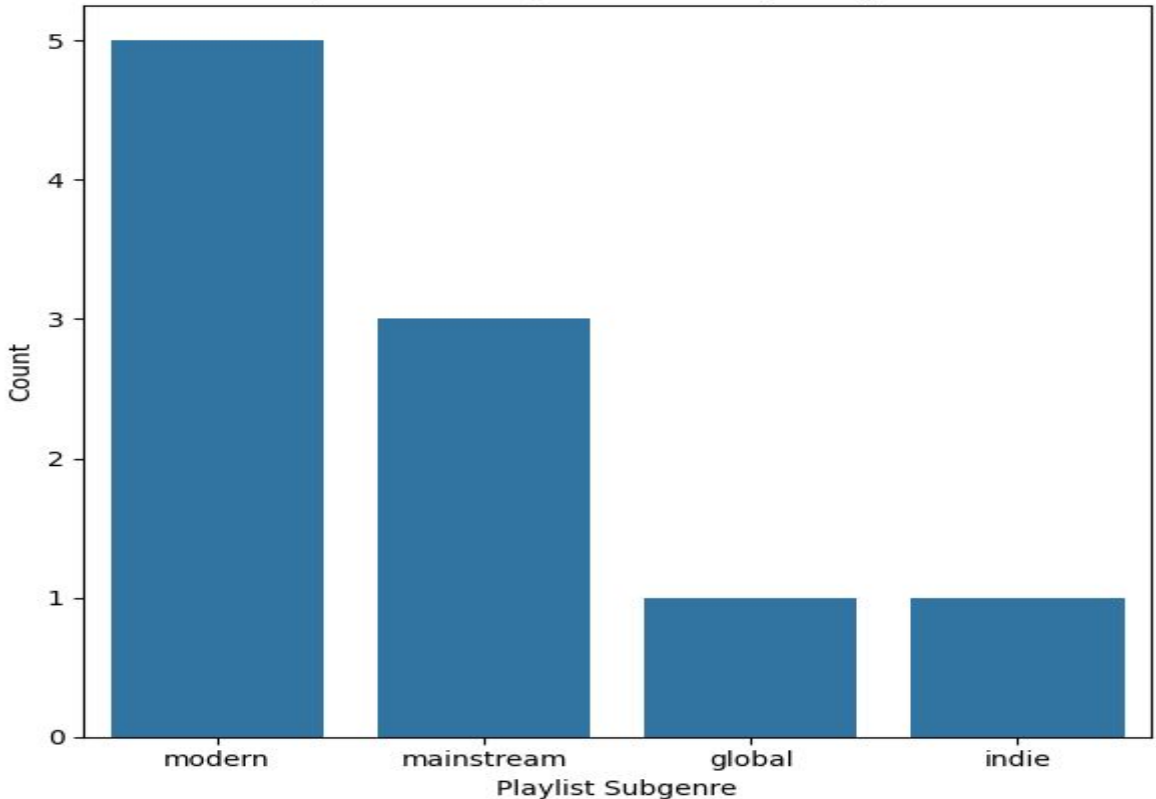
Most popular tracks coming from **pop** , **gaming** playlist genre and **modern mainstream** sub-genre.

Top_10	Track_popularity	Playlist_genre	Playlist_subgenre
0	100	gaming	modern
1	98	pop	mainstream
2	97	gaming	modern
3	94	gaming	modern
4	93	latin	modern
5	93	latin	modern
6	93	pop	global
7	93	pop	mainstream
8	93	folk	indie
9	93	pop	mainstream

Top 10 Most Popular Tracks by Genre

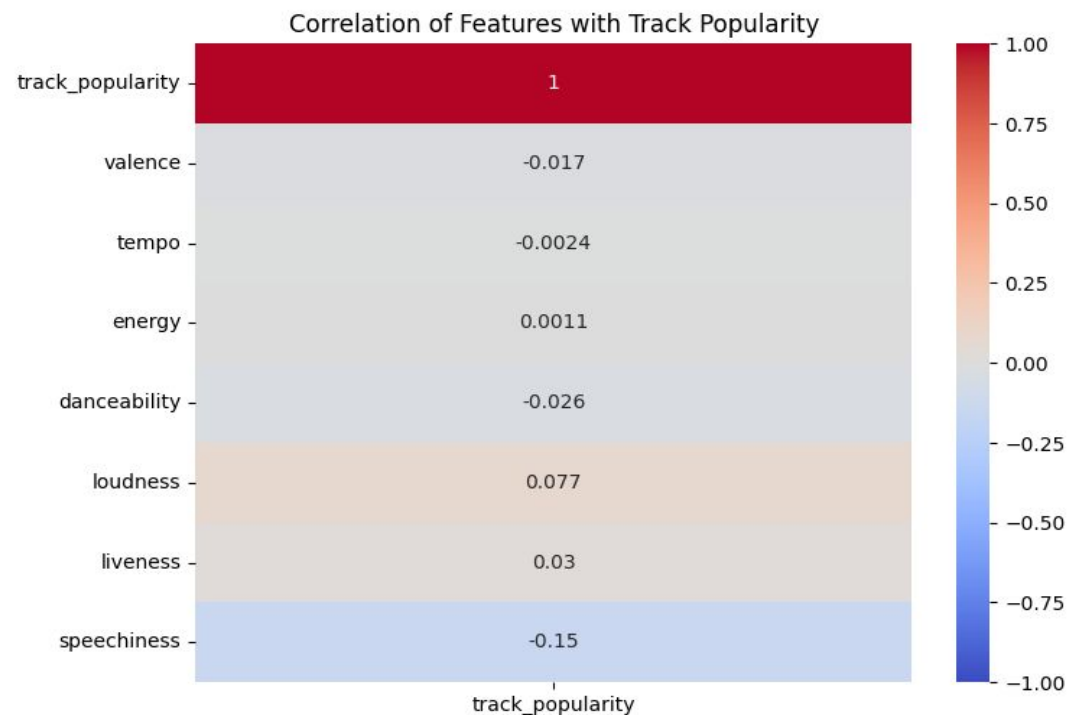


Top 10 Most Popular Tracks by Subgenre

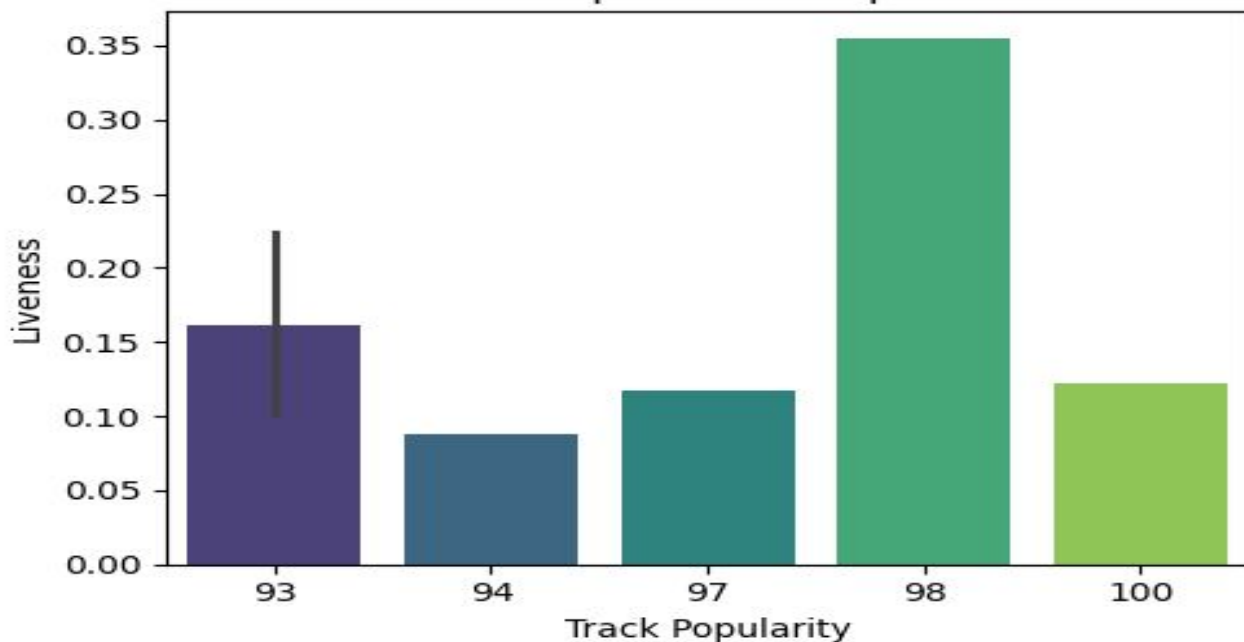


Tracks Popularity

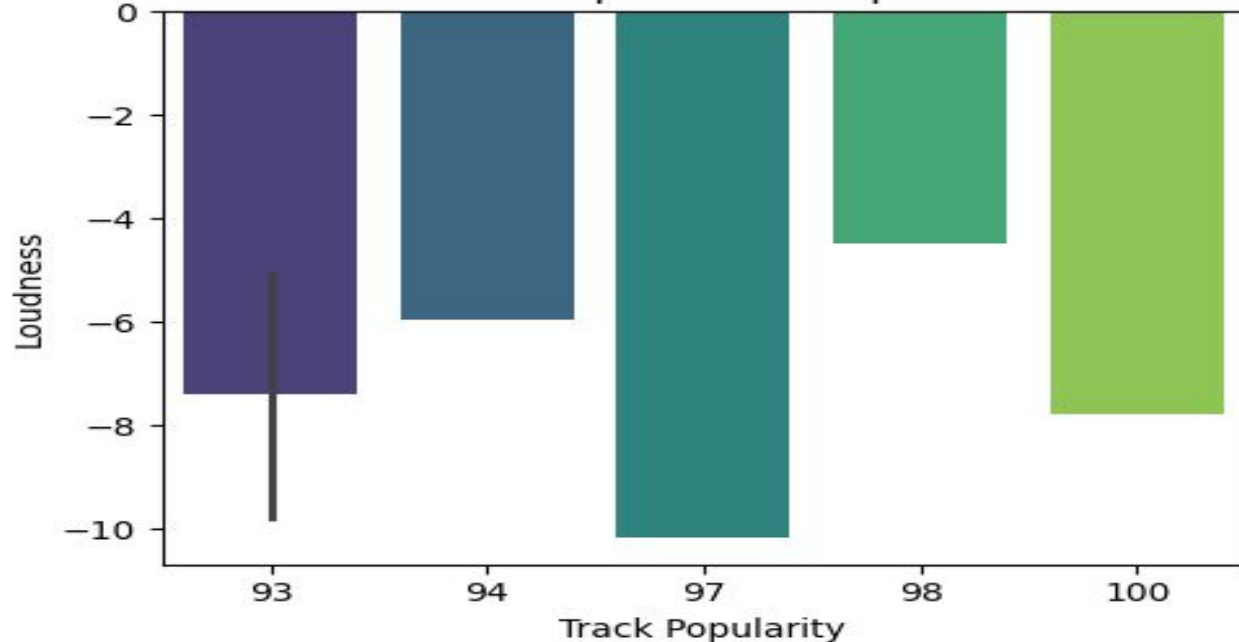
Tracks popularity mostly depends on **loudness** or **liveness** than any other features of tracks.



Liveness of Top 10 Most Popular Tracks

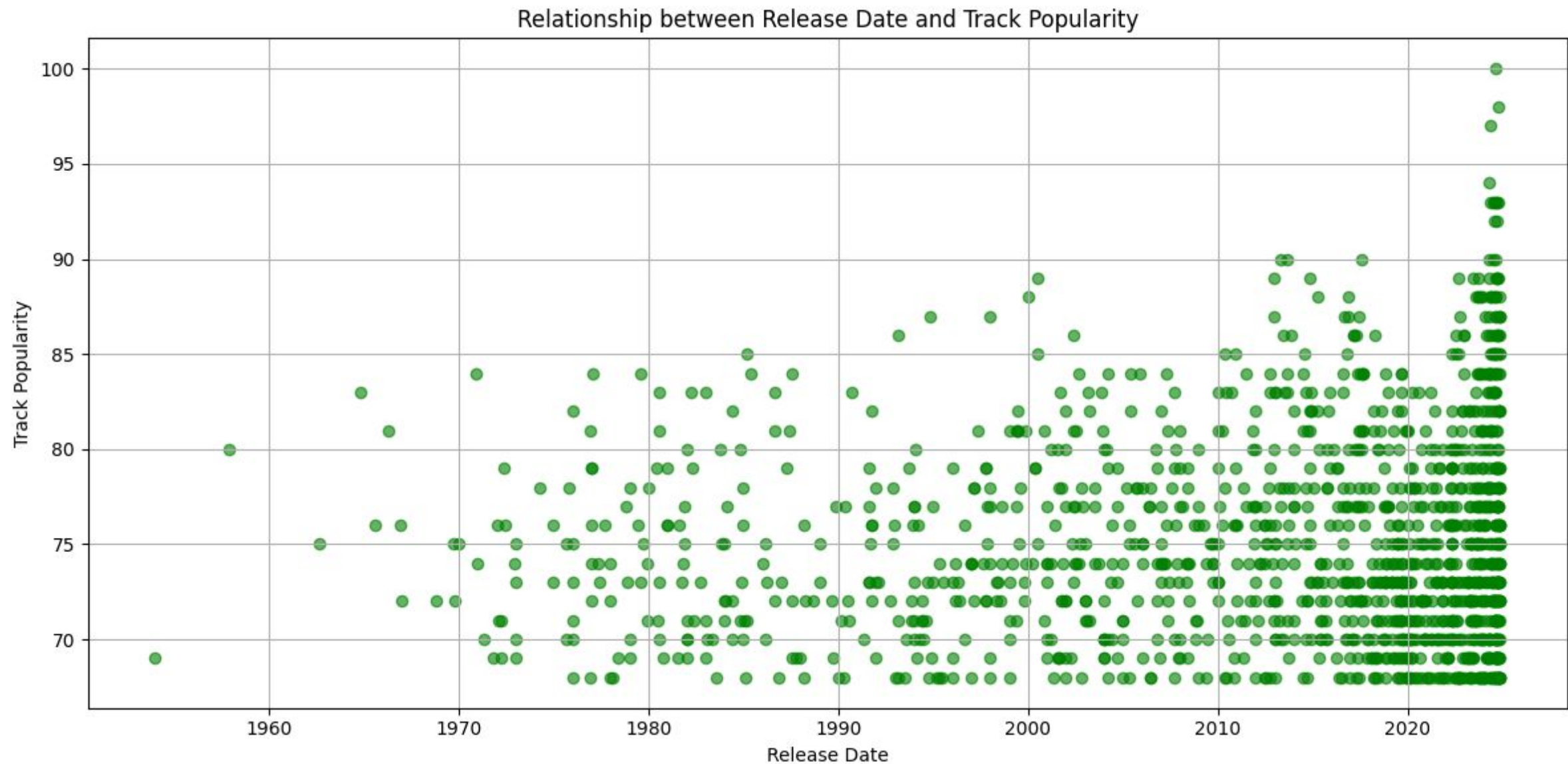


Loudness of Top 10 Most Popular Tracks



Tracks (Released in recent)

Tracks **released in recent years** are showing **higher popularity**, reflecting **listeners'** inclination towards new music.



Key Differences in Features

Valence: Valence measures the overall mood of a track. **Popular playlists might have tracks with higher valence, indicating more happy and upbeat songs.**

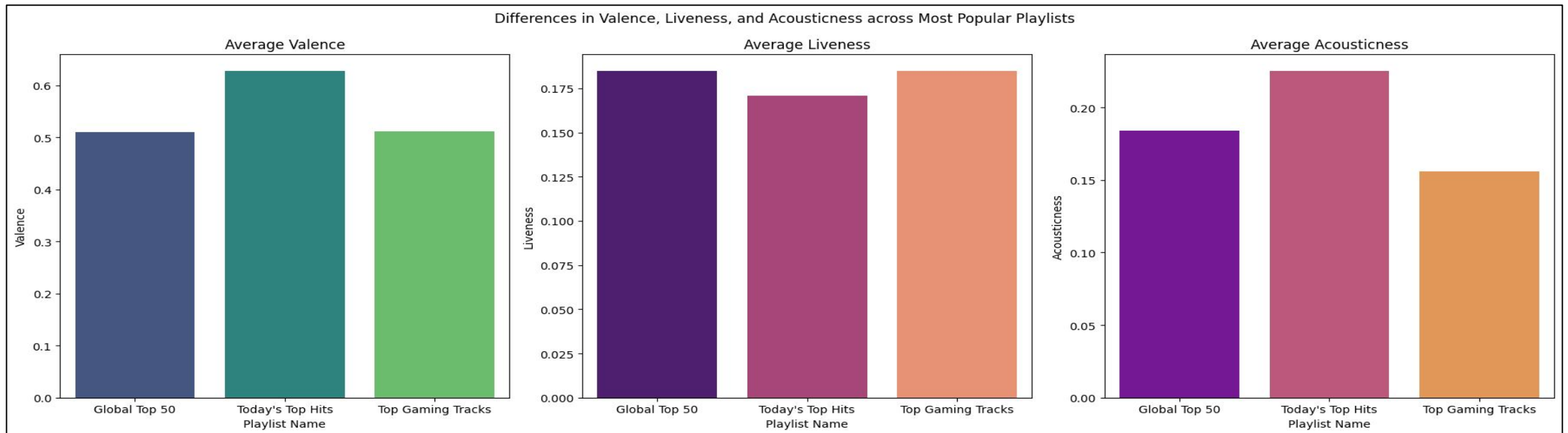
Visualization: A bar plot showing the average valence for each popular playlist.

Liveness: Liveness detects the presence of an audience in the recording. **High liveness values could indicate live performances, which may appeal to certain listeners.**

Visualization: A box plot comparing liveness values across popular playlists.

Acousticness: Acousticness measures how acoustic a track is. **Playlists with higher acousticness may feature more acoustic or unplugged versions of songs.**

Visualization: A histogram showing the distribution of acousticness values in each popular playlist.



Key Result

Insights and Patterns in Popular Tracks

High Energy and Danceability: Popular tracks often have **high energy and danceability**, indicating that **listeners prefer lively and danceable** music.

Genre Trends: Certain genres (**pop, rock**) might consistently **show higher popularity averages**, suggesting genre preferences among Spotify users.

Recent Releases: Tracks **released in recent years might show higher popularity**, reflecting listeners' inclination towards new music.

Playlist Influence: Tracks included in **popular playlists often become more popular themselves**, showing how being featured can boost a track's popularity

Collaborative Impact: Collaborations between **well-known artists can result in higher popularity** due to combined fan bases.

Correlation Insights: Features like **energy and valence are positively correlated**, while **acousticness tends to have a negative correlation with valence**, implying that more acoustic tracks are perceived as less happy.

Implications for Spotify and the Music Industry

Personalization and User Engagement: The insights into user preferences for high-energy and danceable tracks can inform Spotify's recommendation algorithms, enhancing user engagement by tailoring playlists and recommendations to listener tastes.

Genre Focus: Understanding genre popularity helps Spotify and artists focus their marketing and promotional efforts on genres that are more likely to attract listeners, potentially increasing streams and revenue.

Artist Promotion: By recognizing the influence of playlists, Spotify can develop strategies to promote emerging artists through curated playlists, providing them with a platform to reach a broader audience.

Release Strategies: Artists and record labels can leverage the trend towards newer music by strategically timing their releases and aligning them with significant events or promotional campaigns to maximize visibility and streams.

Content Curation: Spotify can continue to refine its curated playlists by incorporating data-driven insights, ensuring they remain relevant and appealing to users, ultimately driving user retention and satisfaction.

Thank you!!!